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FACULTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TD3-Reseau1

Nom : SPADY

Prénom : Jean-Wisly

Niveau : L3

Le 03/11/2026

Objectif du TD

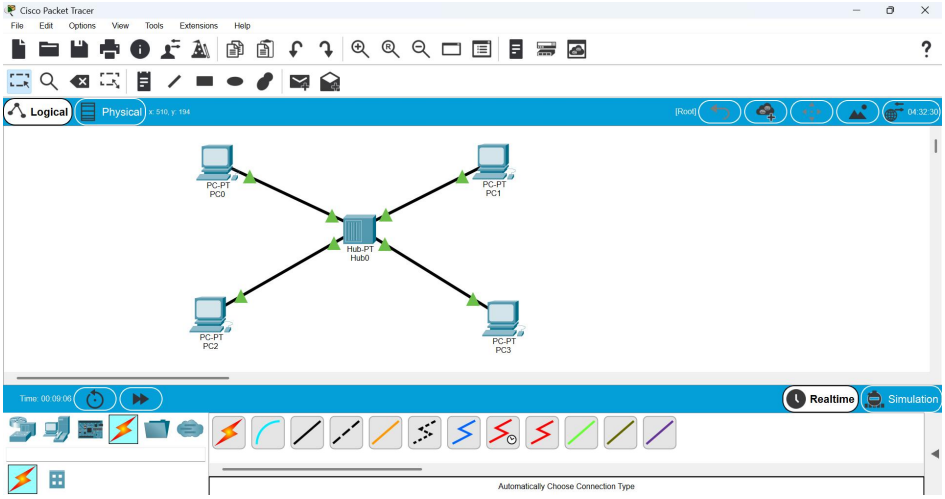
Ce TD m'a permis d`apprendre à construire  des topologies et configurer les adresses IP des différents équipements et vérifier la communication entre les machines à l`aide de la commande ping.

La démarche suivie est :

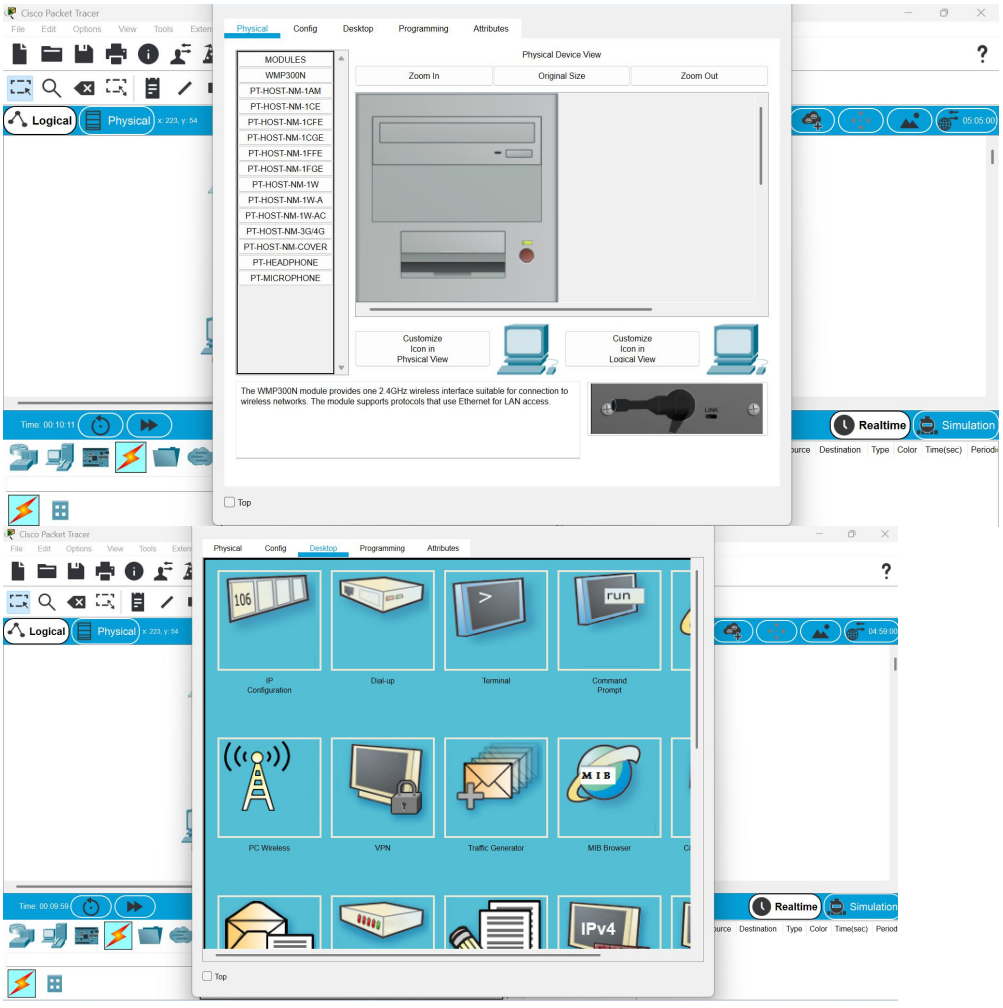
- Ouvrir Cisco Packet tracer
- Créer les différents topologies
- Configurer les adresses IPv4 et IPv6
- Tester la connectivité à l'aide de la commande ping

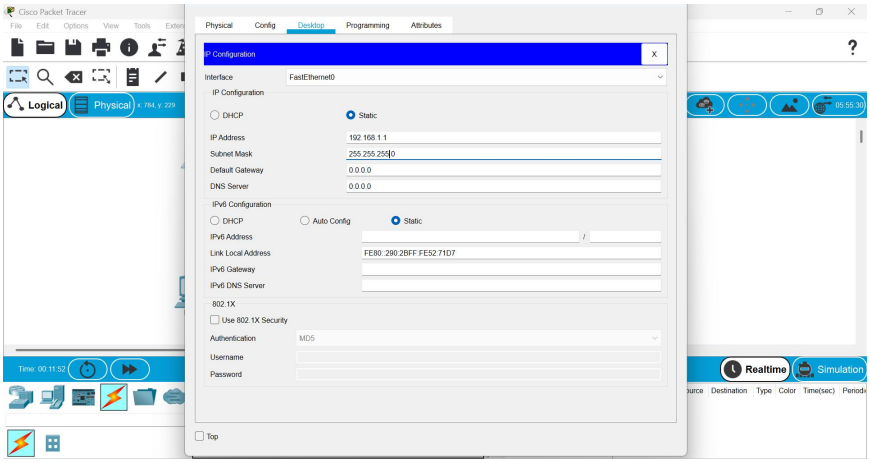
Première partie

1) Créons une topologie avec un Hub et 4 PC connectés au switch

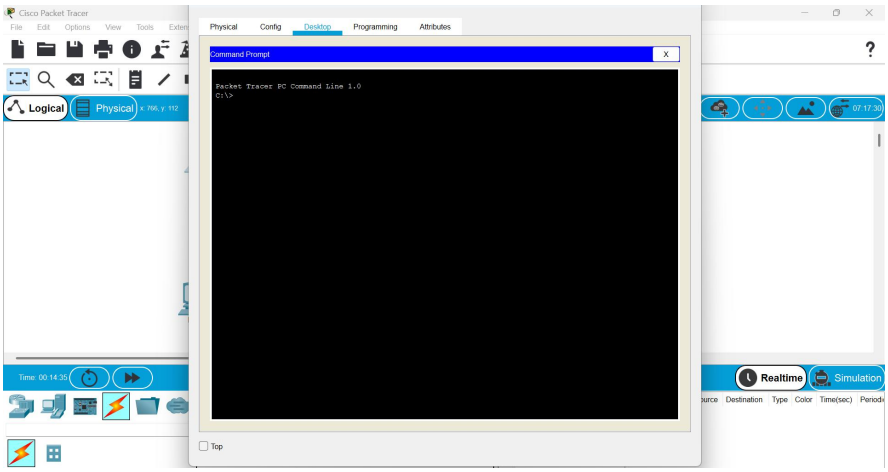


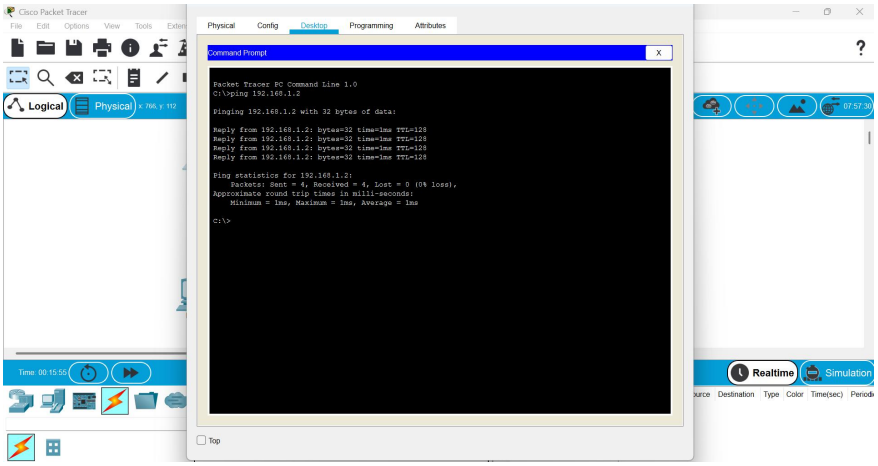
2) Configurons les adresses IPv4



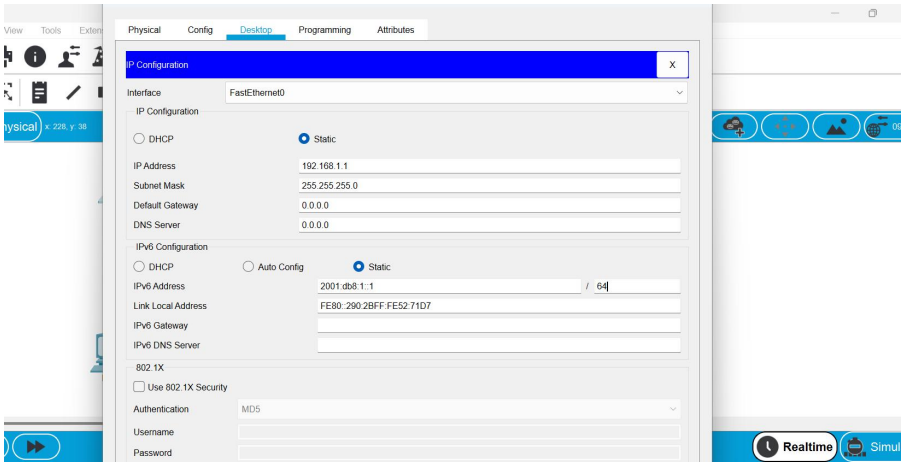


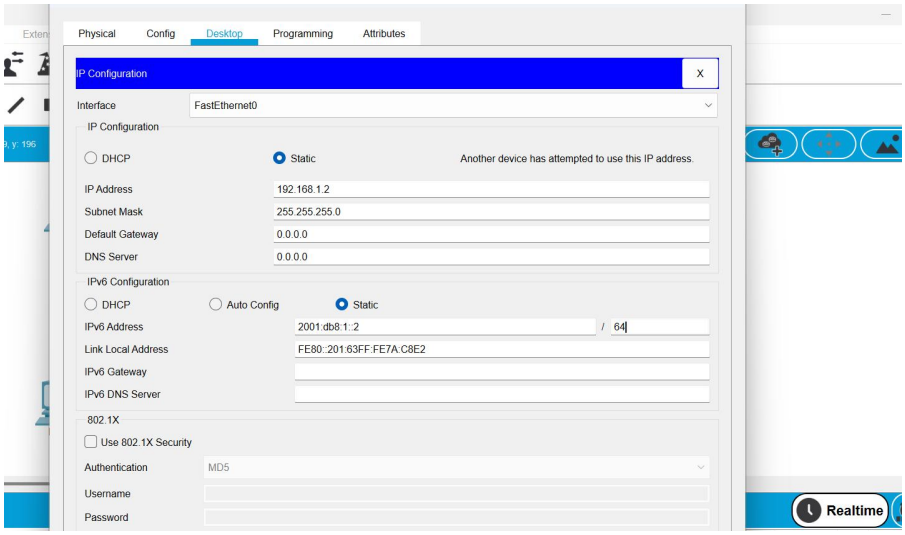
3) Vérifions la connectivité en faisant un ping entre tous les pc

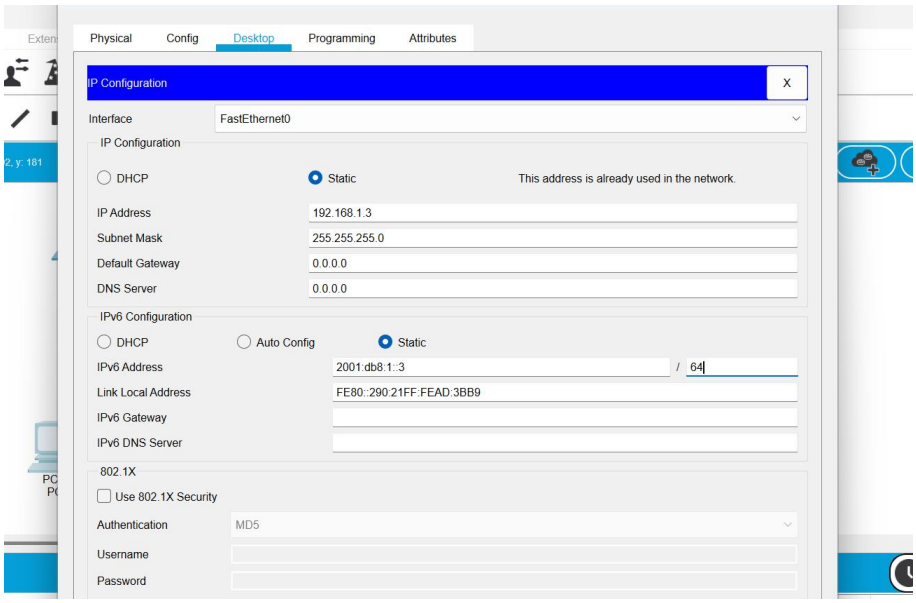


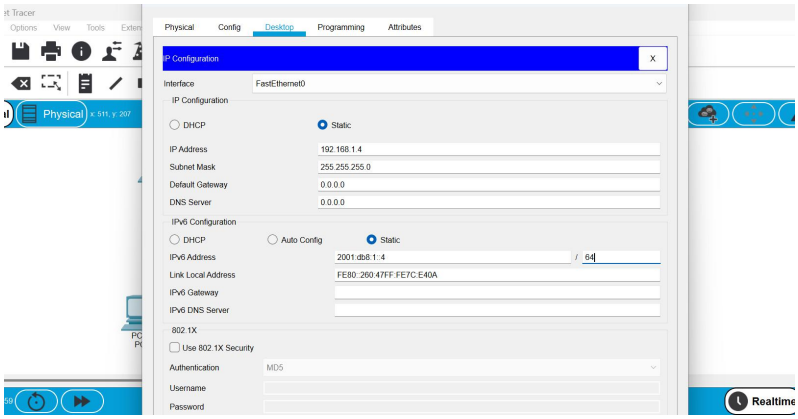


4) Configurons une adresse IPv6 sur chaque PC dans la même topologie


Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
192.168.1.1
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
2001:db8:1::1
/ 64
Link Local Address
FE80::290:2BFF:FE52:71D7
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Realtime


Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
Another device has attempted to use this IP address.
IP Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
2001:db8:1::2
/ 64
Link Local Address
FE80::201:63FF:FE7A:C8E2
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Realtime


Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
This address is already used in the network.
IP Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
2001:db8:1::3
/ 64
Link Local Address
FE80::290:21FF:FEAD:3BB9
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



5) Testons la conectivité en utilisant la commande

```
C:\>ping 2001:db8:1::1

Pinging 2001:db8:1::1 with 32 bytes of data:

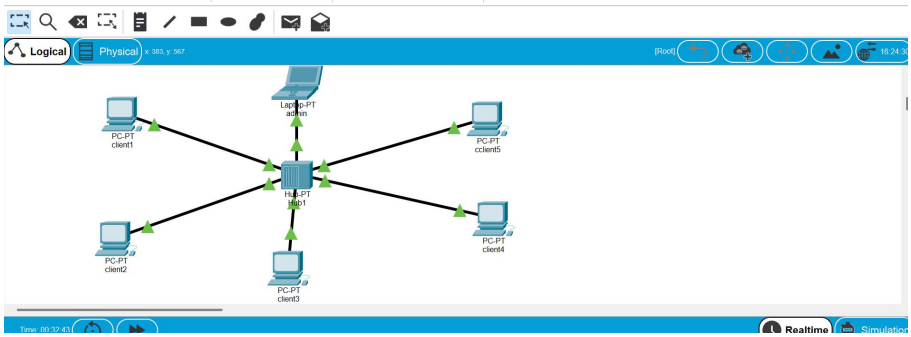
Reply from 2001:DB8:1::1: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=15ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128

Ping statistics for 2001:DB8:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 5ms

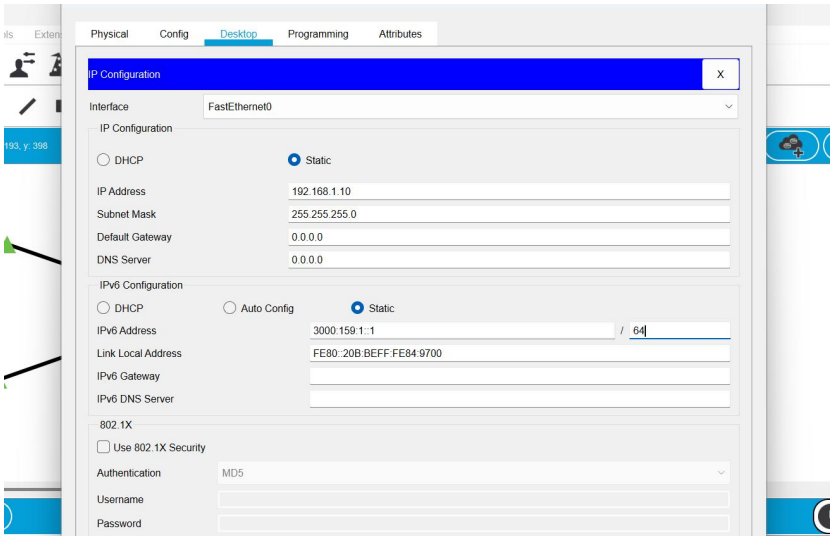
C:\>
```

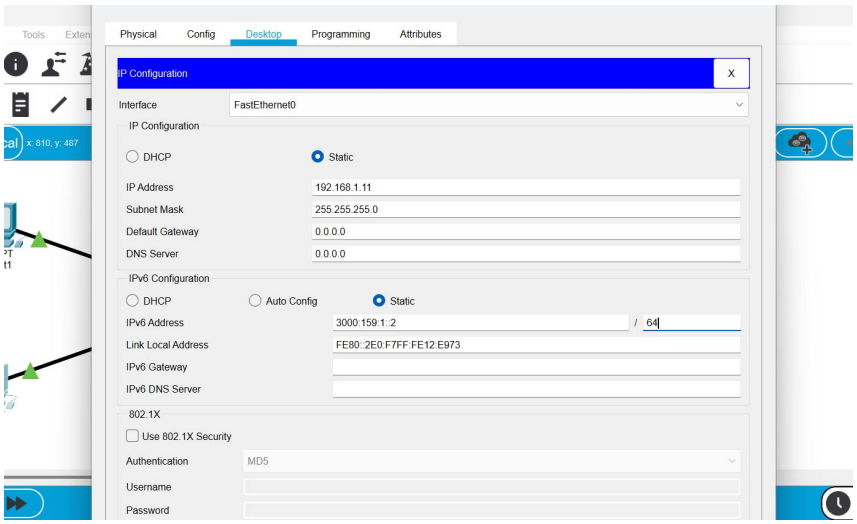
Deuxième partie

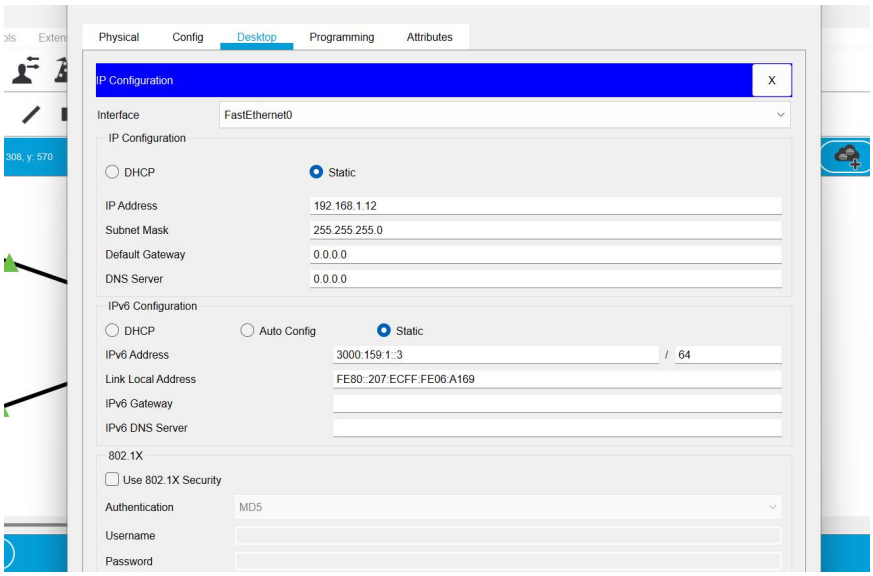
1. La reproduction de la topologie choisie

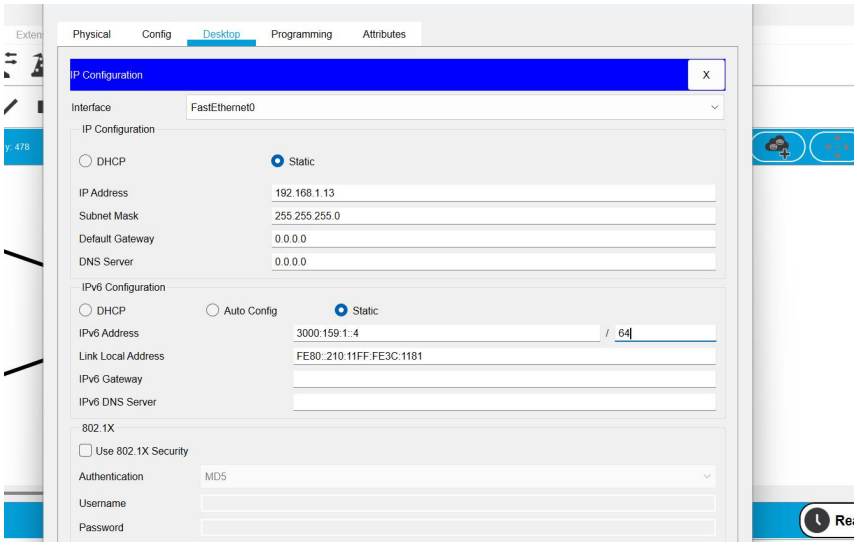


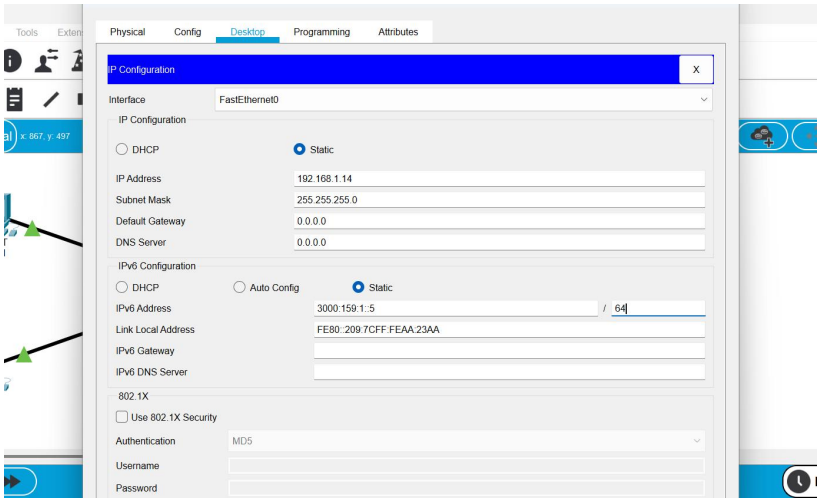
2. Configuration des adresses IPv4 et IPv6


Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
3000:159:1::1
/ 64
Link Local Address
FE80::20B:BEFF:FE84:9700
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

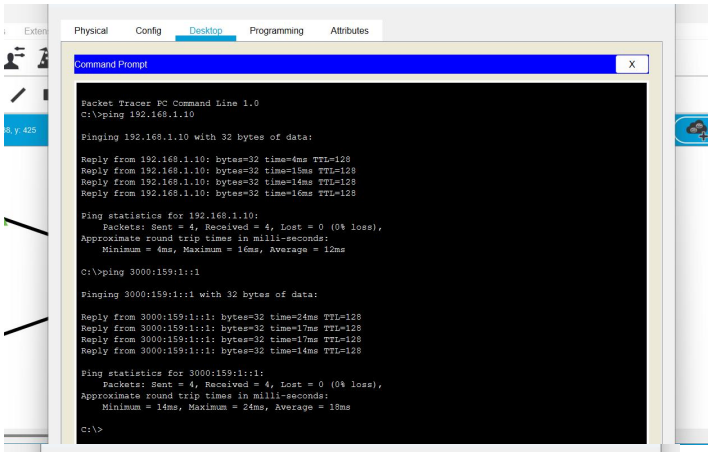

Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
3000:159:1::2
/ 64
Link Local Address
FE80::2E0:F7FF:FE12:E973
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

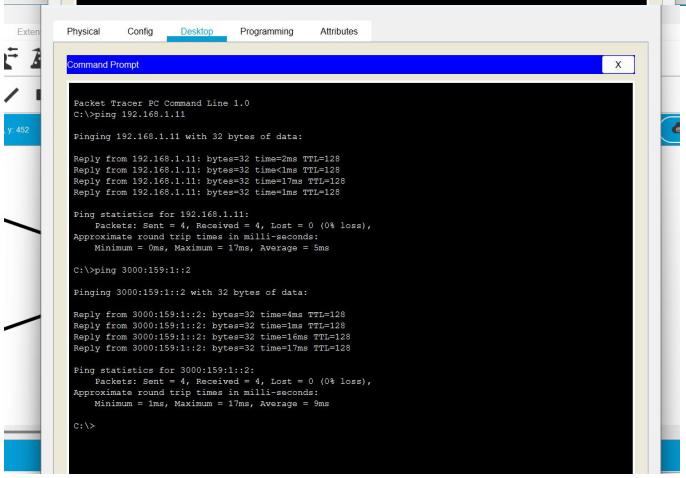

Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
192.168.1.12
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
3000:159:1::3
/ 64
Link Local Address
FE80::207:ECFF:FE06:A169
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

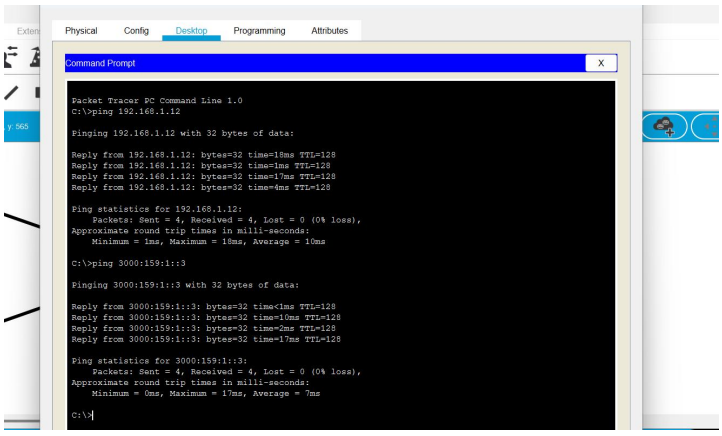


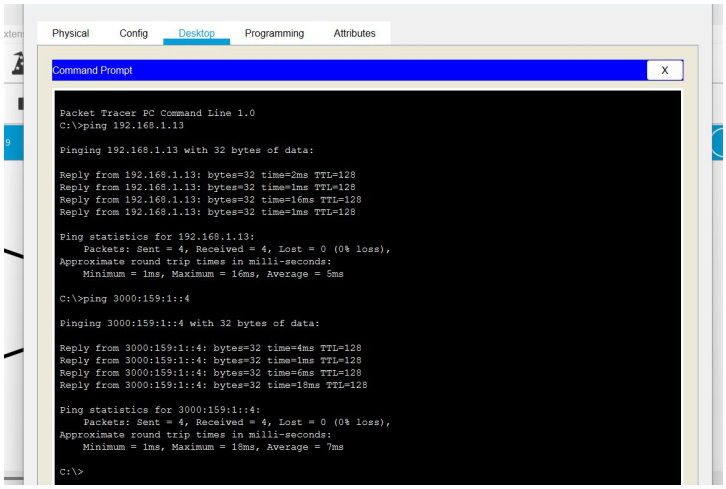


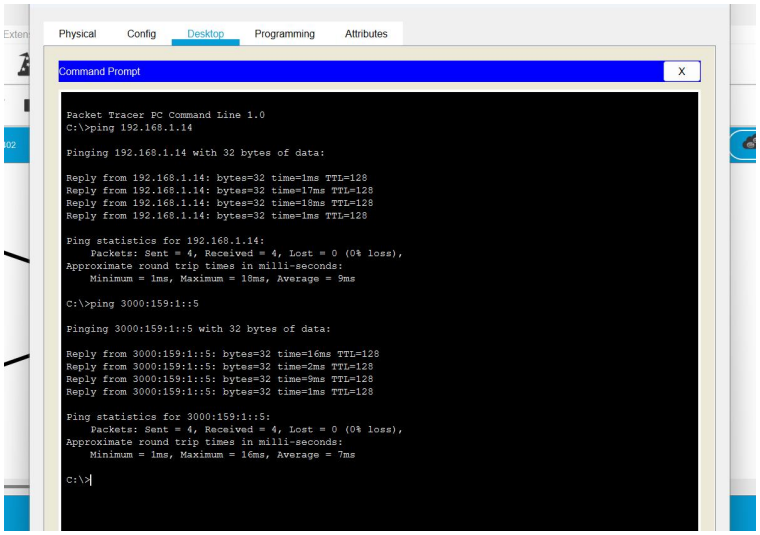
3. Testons la connectivité de l'IPv4 et de l'IPv6 de chaque client


Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=4ms TTL=128
Reply from 192.168.1.10: bytes=32 time=15ms TTL=128
Reply from 192.168.1.10: bytes=32 time=14ms TTL=128
Reply from 192.168.1.10: bytes=32 time=16ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 16ms, Average = 12ms
C:\>ping 3000:159:1::1
Pinging 3000:159:1::1 with 32 bytes of data:
Reply from 3000:159:1::1: bytes=32 time=24ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=17ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=17ms TTL=128
Reply from 3000:159:1::1: bytes=32 time=14ms TTL=128
Ping statistics for 3000:159:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 14ms, Maximum = 24ms, Average = 18ms
C:\>


Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=2ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=17ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 5ms
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=16ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=17ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 17ms, Average = 9ms
C:\>


Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time=18ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=17ms TTL=128
Reply from 192.168.1.12: bytes=32 time=4ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 18ms, Average = 10ms
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=10ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=2ms TTL=128
Reply from 3000:159:1::3: bytes=32 time=17ms TTL=128
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 7ms
C:\>





Conclusion

Ce TD permet de configurer les adresses IP et de vérifier la communication entre les machines à l`aide de la commande Ping.

La diffuculté rencontrée c'est que l'adresse IPv6 "3000:is9:1::x/64" affichait "IPv6 address is invalid" pour résoudre ce problème j'ai changé l'adresse IPv6 qu'on m'a donné en "3000:159:1::x"